3558. Number of Ways to Assign Edge Weights I

Difficulty: Medium

Link: https://leetcode.com/problems/number-of-ways-to-assign-edge-weights-i/description/?envType=daily-question&envId=2026-06-11

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Problem:

There is an undirected tree with `n` nodes labeled from 1 to `n`, rooted at node 1. The tree is represented by a 2D integer array `edges` of length `n - 1`, where `edges[i] = [`u_i`, `v_i`]` indicates that there is an edge between nodes u_i and v_i.

Initially, all edges have a weight of 0. You must assign each edge a weight of either **1** or **2**.

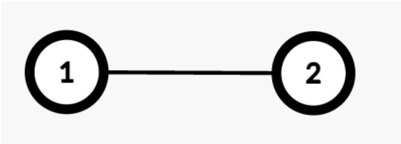
The **cost** of a path between any two nodes `u` and `v` is the total weight of all edges in the path connecting them.

Select any one node `x` at the **maximum** depth. Return the number of ways to assign edge weights in the path from node 1 to `x` such that its total cost is **odd**.

Since the answer may be large, return it **modulo** $10^9 + 7$.

Note: Ignore all edges **not** in the path from node 1 to `x`.

Example 1:



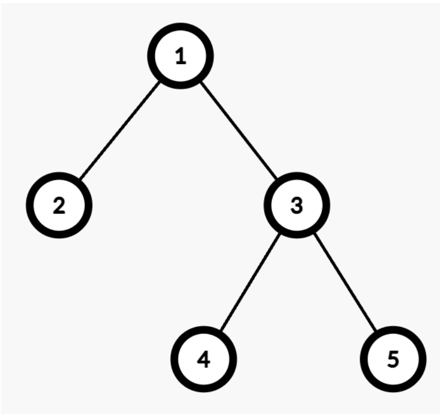
Input: edges = [[1,2]]

Output: 1

Explanation:

- The path from Node 1 to Node 2 consists of one edge (`1 → 2`).
- Assigning weight 1 makes the cost odd, while 2 makes it even. Thus, the number of valid assignments is 1.

Example 2:



Input: edges = [[1,2],[1,3],[3,4],[3,5]]

Output: 2

Explanation:

- The maximum depth is 2, with nodes 4 and 5 at the same depth. Either node can be selected for processing.
- For example, the path from Node 1 to Node 4 consists of two edges (`1 → 3` and `3 → 4`).
- Assigning weights (1,2) or (2,1) results in an odd cost. Thus, the number of valid assignments is 2.

Constraints:

- $2 <= n <= 10^5$
- `edges.length == n - 1`
- $\texttt{edges[i] == [u}_i\texttt{, v}_i\texttt{]}$
- $1 <= u_i, v_i <= n$
- `edges` represents a valid tree.